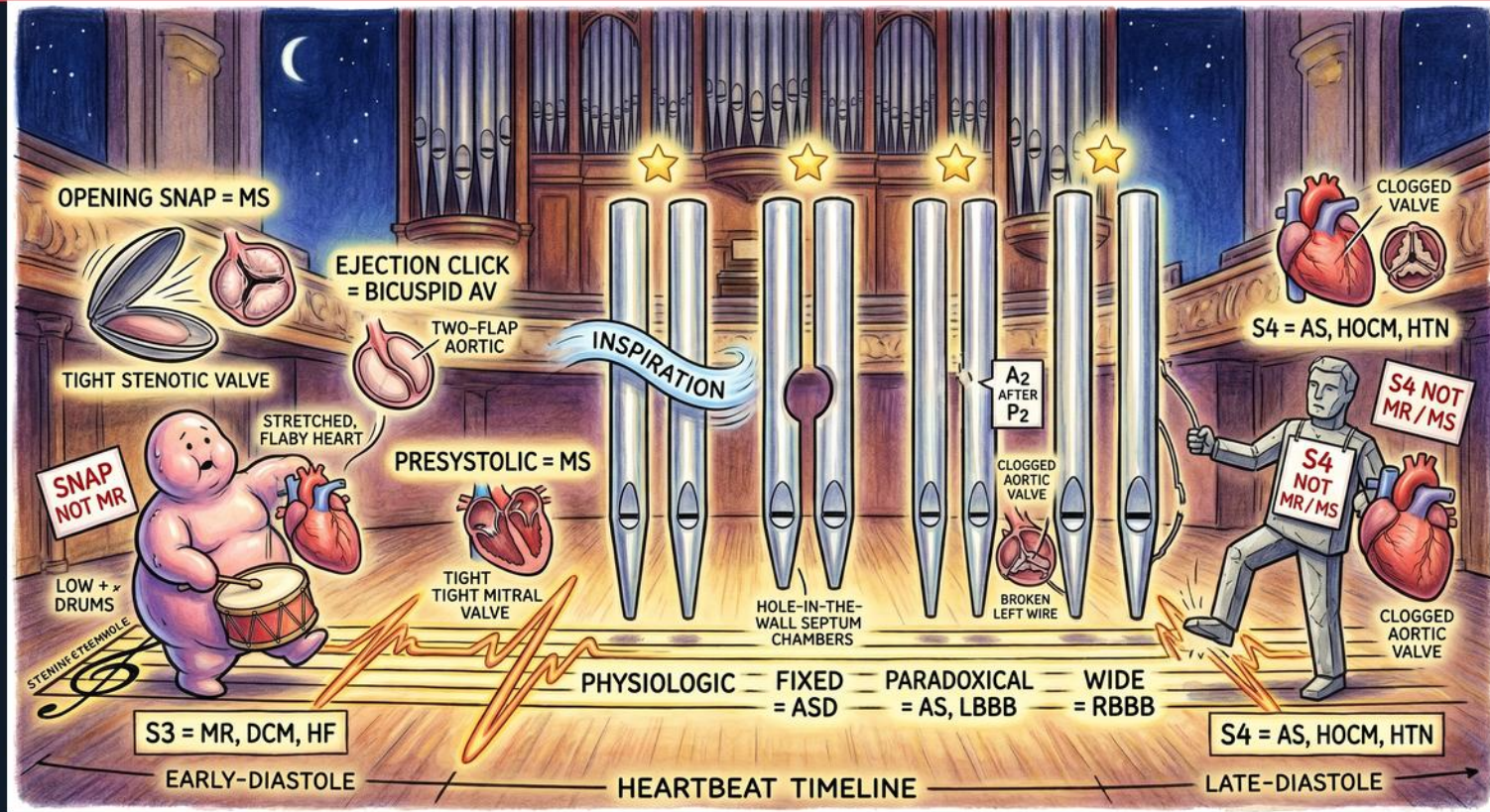


Cardiac Exam — Heart Sounds & S2 Splitting





S3 = MR, DCM, HF

WHAT YOU SEE

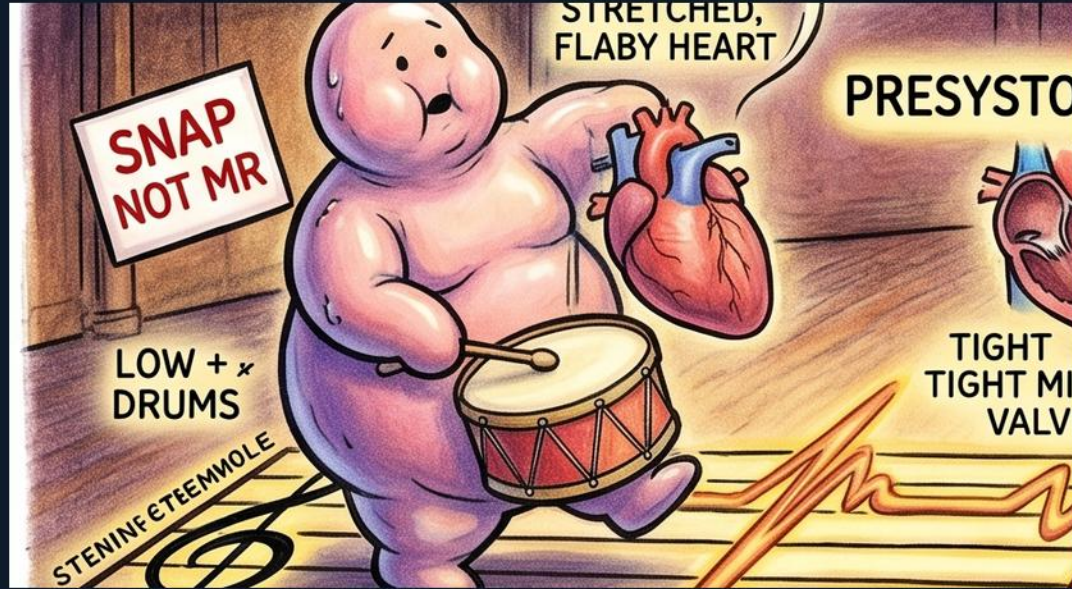
A placard bottom-left 'S3 = MR, DCM, HF' under a marshmallow figure beating a drum, labeled 'EARLY-DIASTOLE'.

WHAT IT ENCODES

S3 is a low-pitched EARLY-DIASTOLIC sound from RAPID VENTRICULAR FILLING into a volume-overloaded ventricle: severe MITRAL REGURGITATION, DILATED CARDIOMYOPATHY, and HEART FAILURE. It can be physiologic in young, healthy people.

THE BOARD TRAP

S3 = volume overload (MR/DCM/HF) and is early diastolic — don't attribute it to a stiff ventricle; that's S4.



Big drum / volume overload

WHAT YOU SEE

The round figure pounding a big drum, with 'LOW + DRUMS' and 'STRETCHED FLABBY HEART'.

WHAT IT ENCODES

The big floppy drum encodes the dilated, volume-loaded, compliant ventricle that generates S3 — blood rushing in during early diastole resonates in the over-filled chamber, giving the low-pitched 'Kentucky' gallop.

THE BOARD TRAP

A big, low, early-diastolic gallop is S3 (floppy/dilated), not the high-pitched late S4 of a stiff ventricle.



Opening snap = MS

WHAT YOU SEE

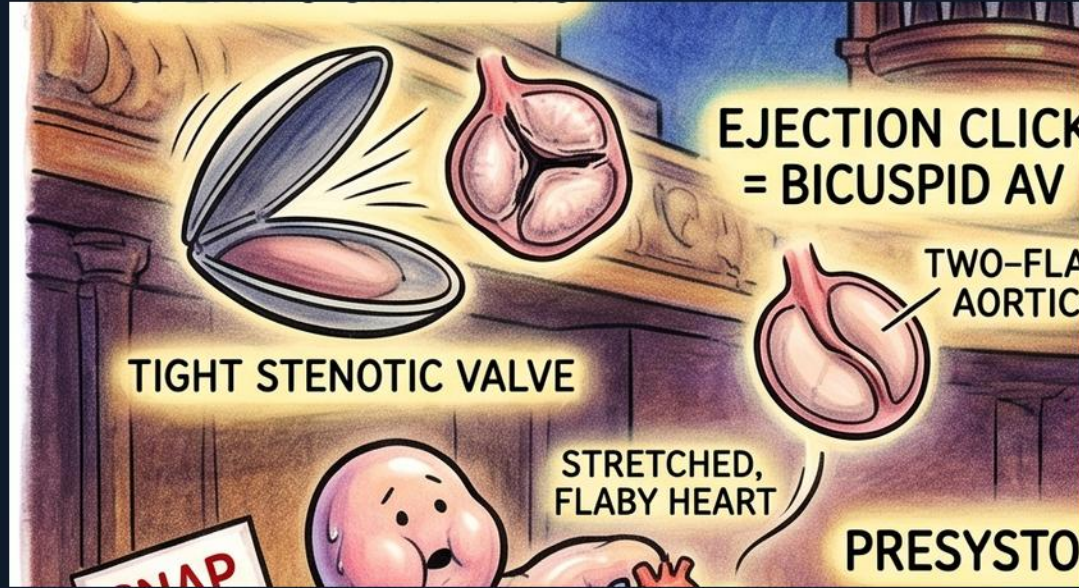
Top-left clam-shell icon labeled 'OPENING SNAP = MS' with 'TIGHT STENOTIC VALVE'.

WHAT IT ENCODES

An OPENING SNAP is a high-pitched EARLY-DIASTOLIC sound from the abrupt opening of a stenotic but still pliable MITRAL valve — the hallmark of MITRAL STENOSIS. A shorter S2–OS interval indicates more severe MS.

THE BOARD TRAP

Opening snap is MITRAL STENOSIS, NOT mitral regurgitation — and it is a diastolic opening sound, not a click.



Snap NOT MR

WHAT YOU SEE

A red placard 'SNAP NOT MR' on the figure's chest.

WHAT IT ENCODES

This rules out mitral regurgitation as the cause of an opening snap. The snap is the sudden checked opening of a fused, stenotic mitral valve (MS); MR produces a holosystolic murmur, not a diastolic snap.

THE BOARD TRAP

The trap answer is MR — an opening snap belongs to MS, not regurgitation.



Ejection click = bicuspid AV

WHAT YOU SEE

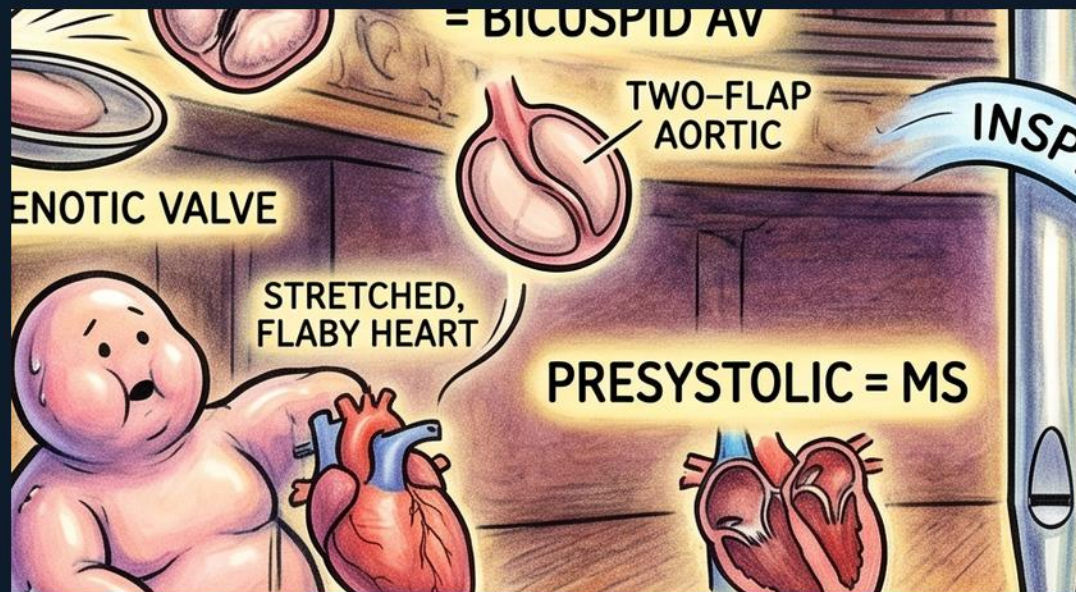
A two-leaflet valve icon labeled 'EJECTION CLICK = BICUSPID AV' with 'TWO-FLAP AORTIC'.

WHAT IT ENCODES

An early-systolic EJECTION CLICK comes from a BICUSPID AORTIC VALVE — the doming of the abnormal two-leaflet valve as it opens. It is often followed by an aortic ejection murmur.

THE BOARD TRAP

An ejection click points to a bicuspid aortic valve (two leaflets), not the diastolic opening snap of MS.



Presystolic = MS

WHAT YOU SEE

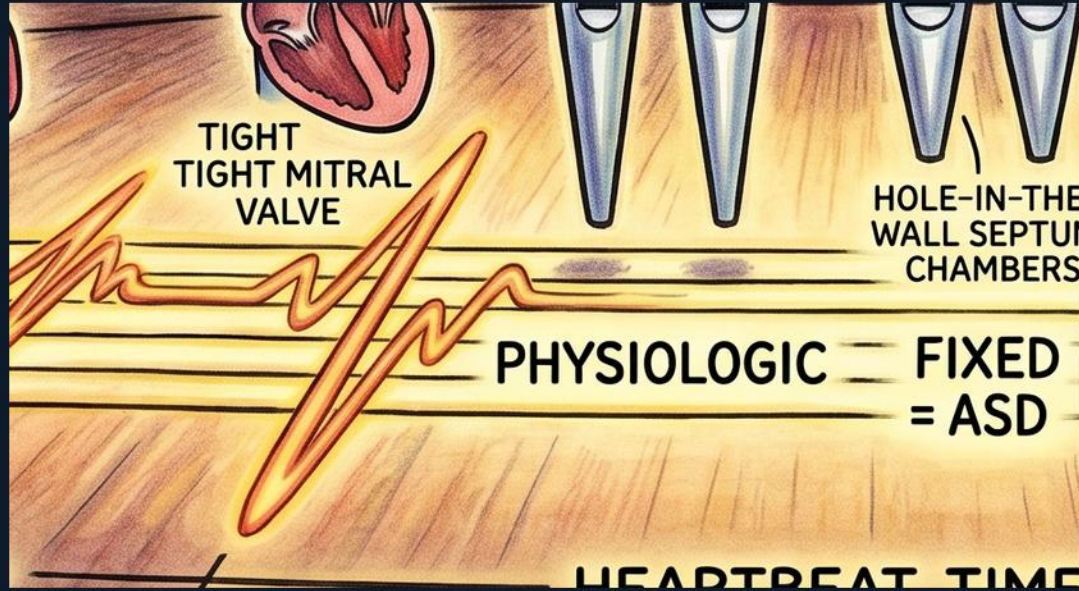
A tight valve icon mid-left labeled 'PRESYSTOLIC = MS' with 'TIGHT MITRAL VALVE'.

WHAT IT ENCODES

A PRESYSTOLIC MURMUR is the late-diastolic accentuation of the MITRAL STENOSIS rumble, caused by atrial contraction pushing blood through the narrowed valve just before S1. It requires sinus rhythm (lost in AFib).

THE BOARD TRAP

Presystolic accentuation is MITRAL STENOSIS — it disappears in atrial fibrillation because it depends on atrial kick.



Physiologic split

WHAT YOU SEE

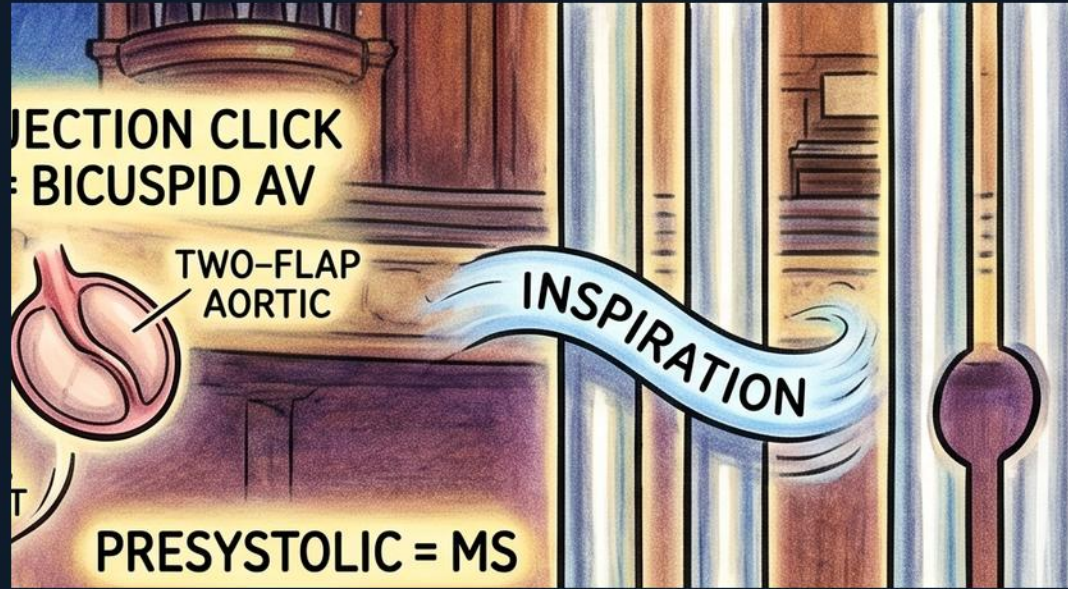
Chimes under the word 'PHYSIOLOGIC' on the heartbeat-timeline label row.

WHAT IT ENCODES

PHYSIOLOGIC splitting of S2 (A2 then P2) WIDENS on INSPIRATION — increased venous return delays pulmonic closure (P2). It narrows on expiration and is a normal finding.

THE BOARD TRAP

Normal split widens with inspiration; a split that doesn't move with respiration (fixed) is abnormal — don't call physiologic splitting pathologic.



Inspiration widens split

WHAT YOU SEE

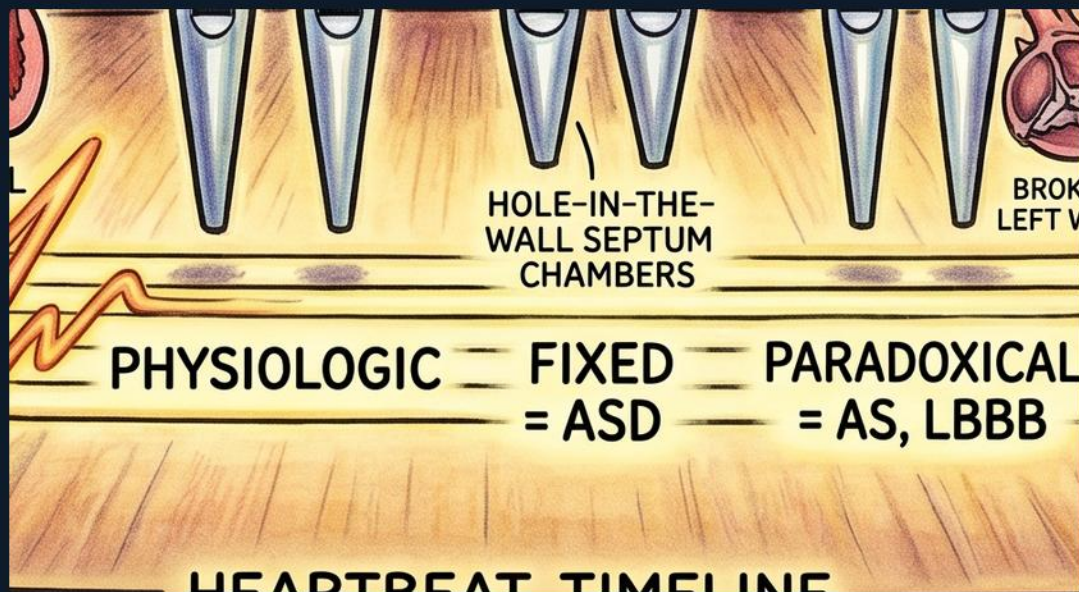
Wind labeled 'INSPIRATION' blowing across the central organ pipes / chimes.

WHAT IT ENCODES

On inspiration, negative intrathoracic pressure increases RV filling and delays pulmonic valve closure, so P2 moves later and the A2-P2 gap widens — the basis of normal physiologic splitting.

THE BOARD TRAP

Respiratory variation is the normal behavior; loss of that variation (fixed split) is the abnormal ASD finding.



Fixed split = ASD

WHAT YOU SEE

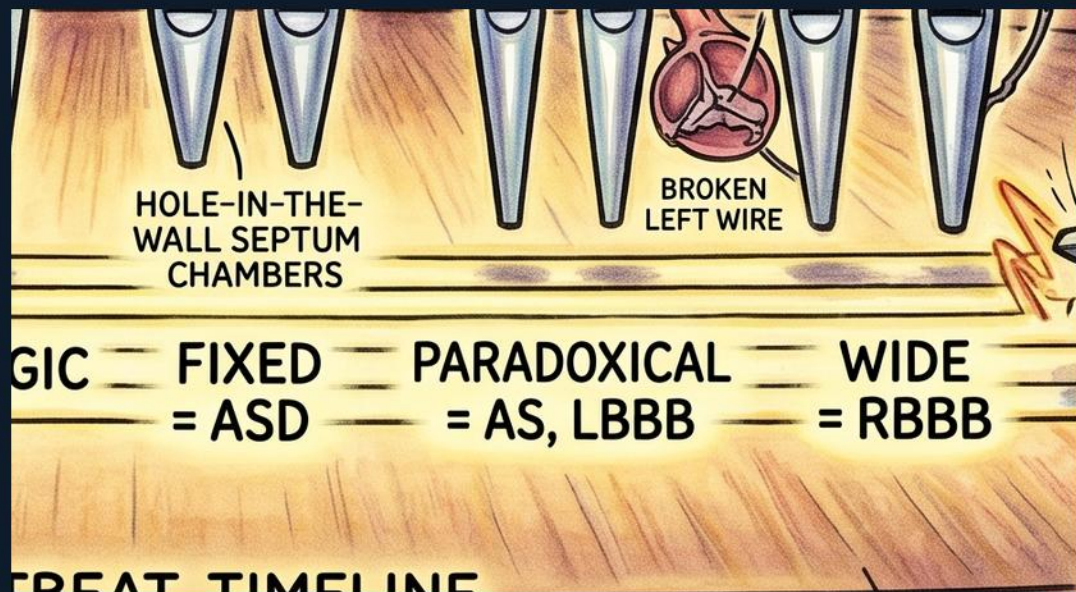
Chimes under the word 'FIXED' labeled 'HOLE-IN-THE-WALL SEPTUM CHAMBERS'.

WHAT IT ENCODES

A FIXED split S2 (A2–P2 gap unchanged through the respiratory cycle) is the classic sign of an ATRIAL SEPTAL DEFECT — the left-to-right shunt equalizes RV loading across respiration, so the split no longer varies.

THE BOARD TRAP

Fixed split = ASD specifically; it does NOT widen with inspiration like a physiologic split.



Paradoxical = AS, LBBB

WHAT YOU SEE

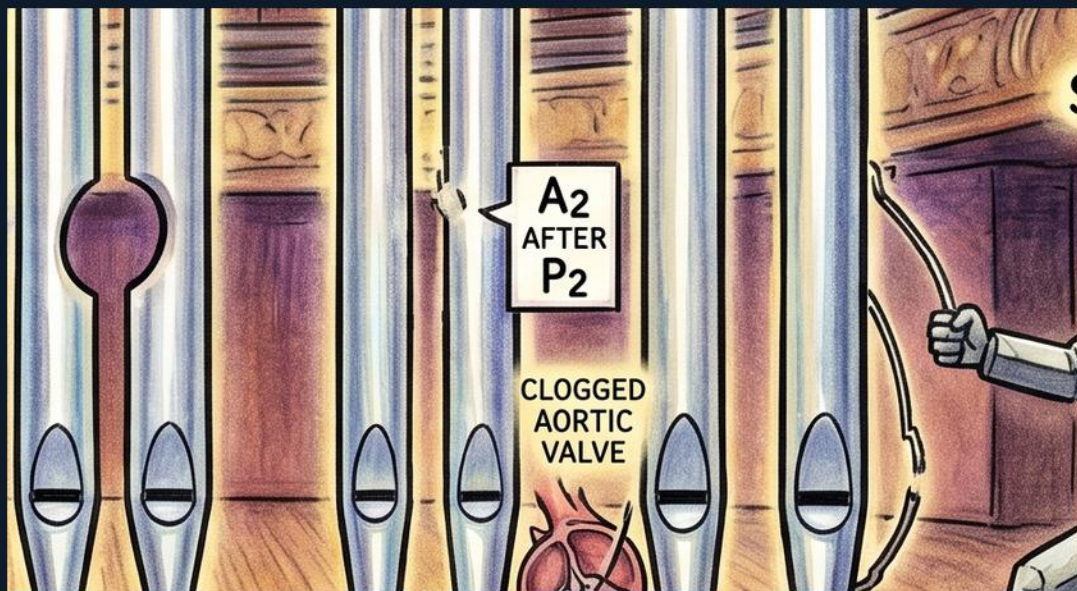
Chimes under 'PARADOXICAL' with note 'A2 AFTER P2' and 'CLOGGED AORTIC VALVE'.

WHAT IT ENCODES

A PARADOXICAL split means A2 comes AFTER P2 (reversed order) because LV emptying is delayed: SEVERE AORTIC STENOSIS or LBBB. The split NARROWS on inspiration (because P2 delays toward the already-late A2).

THE BOARD TRAP

Paradoxical split = A2 AFTER P2 and narrows on inspiration — the reverse of normal; causes are severe AS or LBBB, not RBBB.



A2 after P2

WHAT YOU SEE

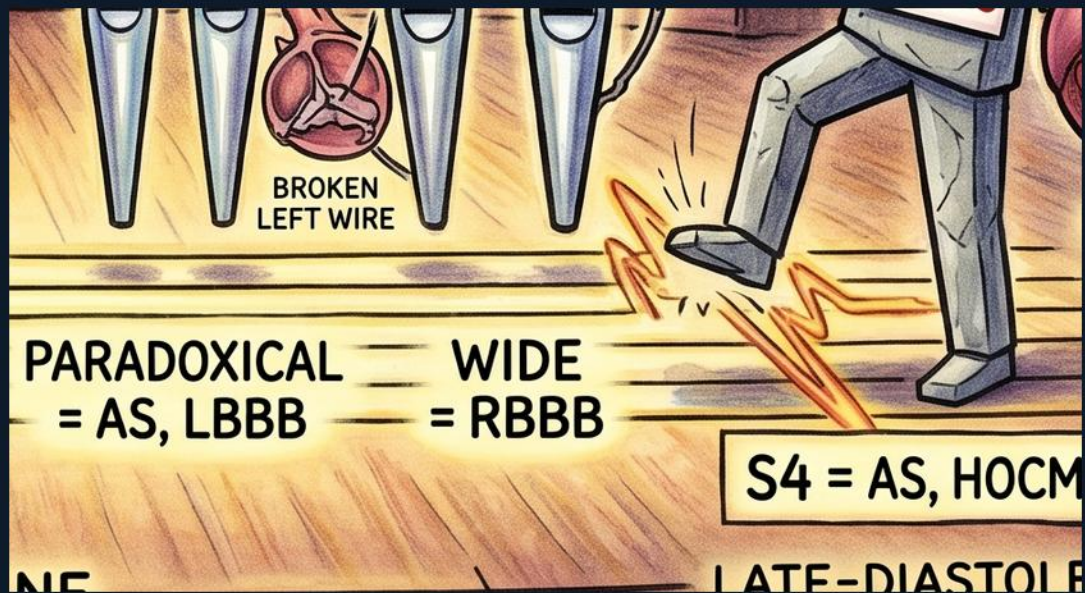
A small placard among the pipes reading 'A2 AFTER P2'.

WHAT IT ENCODES

In paradoxical splitting the AORTIC component is delayed past the pulmonic component, so the normal A2-then-P2 order is reversed. Inspiration delays P2 toward A2, paradoxically narrowing or closing the split.

THE BOARD TRAP

The defining feature is reversed component order (A2 follows P2) — recognizing the reversal is the tested point.



Wide split = RBBB

WHAT YOU SEE

Chimes under 'WIDE' labeled 'BROKEN LEFT WIRE = RBBB' near a snapped wire icon.

WHAT IT ENCODES

A WIDE split S2 (widely separated but still respiration-varying) is caused by RIGHT BUNDLE BRANCH BLOCK — delayed RV activation pushes P2 even later, widening the A2–P2 gap throughout the cycle while still widening further on inspiration.

THE BOARD TRAP

Wide (not fixed) split = RBBB; it still varies with respiration, distinguishing it from the fixed split of ASD.



S4 = AS, HOCM, HTN

WHAT YOU SEE

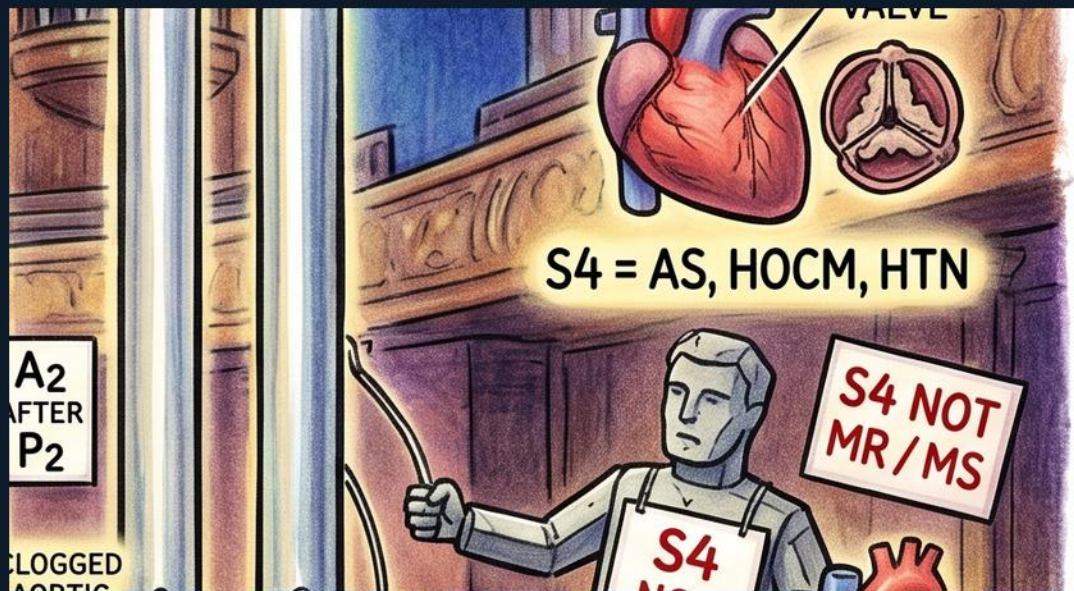
Top-right placard 'S4 = AS, HOCM, HTN' with a clogged/thick valve icon.

WHAT IT ENCODES

S4 is a LATE-DIASTOLIC sound from ATRIAL CONTRACTION (atrial kick) pushing blood into a STIFF, non-compliant ventricle: AORTIC STENOSIS, HOCM, and HYPERTENSION (LVH). It requires organized atrial contraction (absent in AFib).

THE BOARD TRAP

S4 = STIFF ventricle (AS/HOCM/HTN), NOT MR or MS — and it is late diastolic ('Tennessee'), unlike the early-diastolic S3.



S4 NOT MR / MS

WHAT YOU SEE

A red placard 'S4 NOT MR / MS' on the stick-figure pointing right.

WHAT IT ENCODES

This explicitly rules out mitral regurgitation and mitral stenosis as causes of S4. S4 reflects decreased ventricular compliance (stiff LV), not a leaky or stenotic mitral valve.

THE BOARD TRAP

The trap answers are MR and MS — S4 is a stiff-LV sound (AS/HOCM/HTN), not a mitral-valve lesion.



S4 = late diastole / atrial kick

WHAT YOU SEE

Bottom-right placard 'S4 = AS, HOCM, HTN' under 'LATE-DIASTOLE' on the heartbeat timeline.

WHAT IT ENCODES

S4 falls in LATE DIASTOLE, immediately before S1, generated by the atrial kick striking a stiff ventricle. Its timing (late, presystolic) and its stiff-LV causes distinguish it from the early-diastolic S3 gallop.

THE BOARD TRAP

Timing matters: S4 is late diastolic (atrial kick into a stiff LV); placing it in early diastole confuses it with S3.

Summary — every symbol

1. S3 = MR, DCM, HF

S3 is a low-pitched EARLY-DIASTOLIC sound from RAPID VENTRICULAR FILLING into a volume-overloaded ventricle: severe MITRAL REGURGITATION, DILATED CARDIOMYOPATHY, and HEART FAILURE

3. Opening snap = MS

An OPENING SNAP is a high-pitched EARLY-DIASTOLIC sound from the abrupt opening of a stenotic but still pliable MITRAL valve — the hallmark of MITRAL STENOSIS

5. Ejection click = bicuspid AV

An early-systolic EJECTION CLICK comes from a BICUSPID AORTIC VALVE — the doming of the abnormal two-leaflet valve as it opens

7. Physiologic split

PHYSIOLOGIC splitting of S2 (A2 then P2) WIDENS on INSPIRATION — increased venous return delays pulmonic closure (P2)

9. Fixed split = ASD

A FIXED split S2 (A2–P2 gap unchanged through the respiratory cycle) is the classic sign of an ATRIAL SEPTAL DEFECT — the left-to-right shunt equalizes RV loading across respiration, so the split no longer varies.

11. A2 after P2

In paradoxical splitting the AORTIC component is delayed past the pulmonic component, so the normal A2-then-P2 order is reversed

13. S4 = AS, HOCM, HTN

S4 is a LATE-DIASTOLIC sound from ATRIAL CONTRACTION (atrial kick) pushing blood into a STIFF, non-compliant ventricle: AORTIC STENOSIS, HOCM, and HYPERTENSION (LVH)

15. S4 = late diastole / atrial kick

S4 falls in LATE DIASTOLE, immediately before S1, generated by the atrial kick striking a stiff ventricle

2. Big drum / volume overload

The big floppy drum encodes the dilated, volume-loaded, compliant ventricle that generates S3 — blood rushing in during early diastole resonates in the over-filled chamber, giving the low-pitched 'Kentucky' gallop.

4. Snap NOT MR

This rules out mitral regurgitation as the cause of an opening snap

6. Presystolic = MS

A PRESYSTOLIC MURMUR is the late-diastolic accentuation of the MITRAL STENOSIS rumble, caused by atrial contraction pushing blood through the narrowed valve just before S1

8. Inspiration widens split

On inspiration, negative intrathoracic pressure increases RV filling and delays pulmonic valve closure, so P2 moves later and the A2–P2 gap widens — the basis of normal physiologic splitting.

10. Paradoxical = AS, LBBB

A PARADOXICAL split means A2 comes AFTER P2 (reversed order) because LV emptying is delayed: SEVERE AORTIC STENOSIS or LBBB

12. Wide split = RBBB

A WIDE split S2 (widely separated but still respiration-varying) is caused by RIGHT BUNDLE BRANCH BLOCK — delayed RV activation pushes P2 even later, widening the A2–P2 gap throughout the cycle while still widening further on inspiration.

14. S4 NOT MR / MS

This explicitly rules out mitral regurgitation and mitral stenosis as causes of S4